# Week 3 - Practice Exercises

## Please note...

Before you begin to code any of the application below, you should design your solution by outlining your **IPO** and **Object Diagram** on paper.

All your solutions should use instantiable classes to separate the calculations from the user input and output.

Decide what input variable(s) you need, and what data types they should be.

Decide what output variable(s) you need, and what data types they should be.

When you have finished implementation, be sure to **test** your application to ensure the calculations are correct.

## Exercise 1 - Pay Application

Having just started college, Bob has been busy looking for a part-time job to fund his new college social life and after only two weeks of looking he has managed to get a job offer! The job comes with different hours, basic rates of pay and over-time rates so he needs to work out how much money he will earn.

Develop an application that would allow Bob to enter his basic pay rate, his number of basic hours work per week and his number of overtime hours per week. The application should then calculate and display Bobs basic pay for the week, his overtime pay for the week and his total pay including overtime.

Save the instantiable class as **Pay.java** and the App class as **PayApp.java**

Note: The overtime rate is 1.5 times the basic pay rate.

## Exercise 2 - Students Union Application

You have been asked to create an application for the Student Union shop which sells caps for €5, shirts for €10 and hoodies for €20.

Your application should allow the user to input the quantity of each item he/she wants to buy and then calculate and output the total cost.

The App class should be called **StudentsUnionApp.java** and the instantiable class **StudentsUnion.java.**

## Exercise 3 – Interest Application

You have borrowed a sum of money from your sister/brother and they have agreed to give you the money only on the condition that you must pay it back with interest. You agree to pay 3% interest per month that goes by without paying back the money.

Develop an application that allows you to enter a number of months and an amount of money. The application should then calculate the interest owed and add it to the original sum and output the total amount to be repaid.

The App class should be called **InterestApp.java** and the instantiable class **Interest.java**

## Exercise 4 – Calculator Application

You have been asked to create an application for a Calculator.

Your application should allow the user to input two numbers, and as output display the sum and difference of those numbers.

The App class should be called **CalculatorApp.java** and the instantiable class **Calculator.java**